

Question	Marking Guidance	Mark	Comments
09.1	1. Restriction endonucleases/enzymes; 2. (Cut DNA) at specific <u>base</u> sequences/pairs OR (Cut DNA) at recognition/restriction sites;	2 (2 x AO2)	2. Accept 'at palindromic sequences'

Question	Marking Guidance	Mark	Comments
09.2	1. (For) primers; 2. (To produce) a complementary base sequence OR (Primers provide starting sequence) for DNA/taq polymerase OR (Primers) stop (original) DNA strands re-joining;	2 (2 x AO2)	

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09.3	Correct answer of $1.35 \times 10^{16} = 2 \text{ marks}$ OR Correct answer of 1.36×10^{16} (due to rounding at an earlier stage of the calculation) = 2 marks OR Correct answer of $1.4 \times 10^{16} = 2 \text{ marks};$ Incorrect answer but shows $2^{50} = 1 \text{ mark};$	2 (2 x AO2)	Ignore any numbers after 1.35.

Question	Marking Guidance	Mark	Comments
09.4	1. Number of nucleotides/repeats/bases OR Length/mass; 2. (Negative) charge;	2 (2 x AO2)	1. Accept weight for mass. 1. Ignore 'short' on its own. 1. Accept number of base pairs. 1. Ignore density/size. 2. Accept 'polarity'. 2. Reject positive (charge).
